

An Architecture Benchmark for Canine Heart-Size Classification on DogHeart

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Abstract

We benchmark multiple deep learning architectures for three-class canine heart-size classification (Large, Normal, Small) on the DogHeart dataset. Under a shared evaluation framework with architecture-specific training settings, a ResNet18 transfer-learning baseline achieves 76.5% validation accuracy (77.7% macro-F1), while a ConvNeXt-Tiny model reaches 77.5% validation accuracy and 79.4% macro-F1. On the 400-image held-out test split, the ConvNeXt-Tiny model attains 77.25% accuracy under the benchmark evaluation protocol. However, a McNemar exact test on the 200-image validation set does not detect a statistically significant improvement of ConvNeXt-Tiny over ResNet18 ($p = 0.856$, two-sided), consistent with the small validation-set size and the +1.0 percentage-point accuracy difference. The dominant error mode is Large–Normal confusion, accounting for 36 of 45 validation errors, reflecting the clinical ambiguity of the boundary between these classes. We document an exploratory design-space study across eight architectures and multiple augmentation strategies, and we provide all code, checkpoints, and prediction outputs as supplementary material. Our results suggest that classification-only pipelines remain substantially below key-point-supervised methods such as RVT (87.3% test accuracy), while differences among local classification baselines should be interpreted cautiously because architecture choices and training settings are confounded.

1 Introduction

Cardiac disease is a common clinical concern in companion animals, and thoracic radiography remains a widely used first-line imaging modality for cardiac assessment because it is accessible, inexpensive, and non-invasive. The vertebral heart scale (VHS) is a standard radiographic measurement strategy: clinicians measure the long and short axes of the cardiac silhouette and normalize them against thoracic vertebral length. Although VHS provides a standardized quantitative metric, manual measurement is subject to inter-observer variability, positioning artifacts, and reader fatigue.

For canine heart-size classification, recent deep models differ in whether they treat the problem as direct image classification or as anatomical landmark estimation followed by measurement. The Regressive Vision Transformer (RVT) [4] follows the latter route: it predicts VHS-related anatomical landmarks and maps them to categorical heart-size labels through an orthogonal measurement layer. RVT achieves strong performance—85.0% validation R-Accuracy and 87.3% test R-Accuracy—but requires additional key-point supervision and a specialized regression head. A simple CNN baseline for DogHeart reports 72.0% accuracy [2], providing a lightweight reference point below the stronger transfer-learning baselines.

This paper addresses the task of three-class dog heart-size classification (Large, Normal, Small) with two primary objectives. First, we establish a reproducible baseline framework covering data loading, training, evaluation, and inference. Second, we investigate how archi-

texture choice, data augmentation, and regularization affect classification performance on this small veterinary dataset.

Our contributions are:

- A complete, documented, and reproducible PyTorch pipeline for the DogHeart classification task, covering training, evaluation, inference, and output validation.
- Systematic evaluation of six architecture families (custom CNN, ResNet, EfficientNetV2, MobileNetV3, RegNet, and ConvNeXt) with uniform validation reporting.
- Empirical finding that strong augmentation, MixUp, label smoothing, and class-balanced loss applied to a ConvNeXt Tiny backbone yield 77.5% validation accuracy and 79.4% macro-F1 under this exploratory protocol.
- A transparent comparison with published DogHeart baselines, clearly distinguishing locally measured results from literature-reported values, with explicit acknowledgment of limitations.

2 Related Work

2.1 Deep Learning for Radiographic Classification

Medical-image datasets are often too small to train large models from scratch, making transfer learning a practical default for radiographic classification. Residual connections [3] support stable optimization of deeper CNNs, EfficientNet’s compound scaling [9] balances width, depth, and resolution, and ConvNeXt modernizes convolutional backbones using design choices inspired by vision transformers [5]. These architecture families provide complementary baselines for DogHeart classification.

In veterinary radiograph analysis, transfer learning is particularly attractive because labeled datasets are often much smaller than natural-image benchmarks. ImageNet-pretrained backbones provide generic low- and mid-level visual features, while the final classifier layers can be adapted to the clinical target. The DogHeart benchmark

studied by RVT [4] is therefore a useful test case for comparing simple classification-only transfer learning with anatomically informed key-point regression.

2.2 Dog Cardiomegaly Assessment

The RVT framework by Li and Zhang [4] represents the current state of the art in automated dog cardiomegaly assessment. Their method employs a Vision Transformer backbone trained with two complementary supervision signals: categorical cross-entropy for heart-size classification (C-Accuracy) and key-point regression for VHS measurement (R-Accuracy). The key-point head predicts 14 anatomical landmarks along the cardiac border, which are then projected to orthogonal axes through a learnable layer. This design explicitly encodes clinical domain knowledge, making the model’s predictions more interpretable to veterinarians.

Table 1 reproduces the commonly cited external comparison results from RVT. Several conventional CNN architectures—including ResNet50 (80.0%), NasNet-Large (80.0%), EfficientNetB7 (82.0%), and CONVT (82.0%)—achieve competitive R-Accuracy values in the RVT protocol. We emphasize that these values are not local reruns but rather literature-reported numbers that provide contextual positioning.

Table 1: External DogHeart comparison values reported by the RVT paper [4]. Values are percentages under the cited paper’s protocol and serve as literature baselines.

Method	Valid C-Acc.	Valid R-Acc.	Test C-Acc.	Test R-Acc.
AlexNet	76.5	77.5	72.0	73.8
GoogleNet	77.0	77.5	71.3	74.8
VGG16	78.5	78.5	74.8	75.0
VGG19	79.5	80.0	76.0	77.3
ResNet50	78.5	80.0	77.3	78.3
NasNetLarge	82.5	80.0	82.3	82.5
EfficientNetB7	81.5	82.0	83.0	84.5
CONVT	81.5	82.0	83.0	85.3
RVT	82.0	85.0	83.3	87.3

2.3 Advanced Training Techniques

Several training techniques have proven effective for improving generalization in limited-data medical imaging scenarios. RandAugment [1] introduces a parameterized

augmentation policy search space that uniformly samples from a set of image transformations. MixUp [10] constructs convex combinations of training examples and their labels, acting as a strong regularizer. Label smoothing [8] replaces hard one-hot targets with softened distributions, reducing overconfidence.

3 Materials and Methods

3.1 Dataset

The DogHeart dataset is organized into training, validation, and test splits. The labeled splits contain images organized into three class folders—Large, Normal, and Small—corresponding to canine heart sizes. We emphasize an important implementation detail: `torchvision.datasets.ImageFolder` sorts class names alphabetically, producing the mapping Large=0, Normal=1, Small=2. This mapping is explicitly fixed in all scripts and stored in every checkpoint.

Table 2 summarizes the dataset statistics. The training set exhibits substantial class imbalance, with the Small class containing only 208 samples compared with 619 for Large and 573 for Normal. This 3:1 imbalance ratio poses challenges for standard training procedures. The validation set preserves a similar class distribution (76 Large, 91 Normal, 33 Small), while the test set contains 400 unlabeled images without class subdirectories.

Table 2: DogHeart dataset statistics. Training data is imbalanced with a 3:1 ratio between the largest and smallest classes.

Split	Large	Normal	Small	Total
Train	619	573	208	1400
Valid	76	91	33	200
Test	–	–	–	400

3.2 Preprocessing and Data Augmentation

All images are converted to RGB tensors and resized to a fixed square resolution ($H \times W$) using bilinear interpolation with anti-aliasing. We evaluate three augmentation strategies with increasing strength:

- **Standard:** Random horizontal flip ($p=0.5$), rotation ($\pm 8^\circ$), and mild color jitter (brightness ± 0.12 , contrast ± 0.12).
- **RandAugment:** Random horizontal flip followed by RandAugment [1] with $N = 2$ operations and magnitude $M = 9$.
- **Strong:** Combines horizontal flip, rotation ($\pm 10^\circ$), color jitter, and RandAugment with $N = 2$, $M = 7$.

All augmentation pipelines conclude with ImageNet-style normalization (mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]). Validation and test inference use deterministic resizing without augmentation.

3.3 MixUp Regularization

We additionally apply MixUp [10] during training, a data-agnostic regularization strategy that constructs virtual training examples by linear interpolation:

$$\tilde{\mathbf{x}} = \lambda \mathbf{x}_i + (1 - \lambda) \mathbf{x}_j, \quad \tilde{\mathbf{y}} = \lambda \mathbf{y}_i + (1 - \lambda) \mathbf{y}_j, \quad (1)$$

where $\mathbf{x}_i, \mathbf{x}_j$ are randomly sampled training images, $\mathbf{y}_i, \mathbf{y}_j$ are their one-hot labels, and $\lambda \sim \text{Beta}(\alpha, \alpha)$. We use $\alpha = 0.2$ for standard runs and $\alpha = 0.4$ for the final ConvNeXt Tiny configuration.

$$\mathcal{L}_{\text{mixup}} = \lambda \mathcal{L}(\hat{\mathbf{y}}, \mathbf{y}_i) + (1 - \lambda) \mathcal{L}(\hat{\mathbf{y}}, \mathbf{y}_j). \quad (2)$$

3.4 Model Architectures

We evaluate a diverse set of architectures spanning different design paradigms and capacity levels:

Custom Residual CNN. A compact network with seven residual blocks (channel progression: $32 \rightarrow 48 \rightarrow 64 \rightarrow 96 \rightarrow 128 \rightarrow 192 \rightarrow 256 \rightarrow 320$) using SiLU activations, batch normalization, and stride-2 downsampling at three stages, followed by global average pooling, dropout ($p = 0.25$), and a fully connected classifier head.

Transfer-Learning CNNs. We adopt ImageNet-pretrained models from torchvision with modified classifier heads. The final fully connected layer is replaced with dropout ($p = 0.25$) followed by a 3-class linear layer.

Table 3 lists the evaluated architectures with their parameter counts.

Table 3: Evaluated architectures.

Architecture	Params (M)
Custom Residual	3.4
ResNet18	11.2
ResNet50	25.6
EfficientNetV2-S	21.5
MobileNetV3-Large	5.5
RegNetY-400MF	4.3
ConvNeXt Tiny	28.6
ConvNeXt Small	50.2

3.5 Loss Function and Optimization

The primary training objective is cross-entropy with label smoothing. For a sample with smoothed target y_c^{sm} and predicted probability p_c :

$$\mathcal{L}_{\text{CE}} = - \sum_{c=1}^3 y_c^{\text{sm}} \log p_c, \quad (3)$$

$$y_c^{\text{sm}} = (1 - \varepsilon) \cdot \mathbb{I}[c = y] + \frac{\varepsilon}{3}, \quad (4)$$

where $\varepsilon = 0.1$ for the final ConvNeXt Tiny configuration. We additionally employ class-balanced weighting in the loss to counteract the training set imbalance:

$$w_c = \frac{N}{3 \cdot N_c}, \quad (5)$$

where N is the total training count and N_c is the count for class c .

The final ConvNeXt Tiny model is optimized with AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$), cosine annealing from $\eta_{\text{max}} = 2 \times 10^{-4}$, weight decay 0.1, batch size 48, label smoothing $\varepsilon = 0.1$, and MixUp $\alpha = 0.4$. Automatic mixed precision (AMP) with `torch.float16` is used for GPU-based experiments.

3.6 Evaluation Metrics

The primary metric is classification accuracy:

$$\text{Accuracy} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}[\hat{y}_i = y_i]. \quad (6)$$

Due to class imbalance, we also report macro-averaged F1-score:

$$\text{Macro-F1} = \frac{1}{3} \sum_{c=1}^3 \frac{2 \cdot \text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}. \quad (7)$$

Per-class precision, recall, and F1 are reported for all experiments. The best checkpoint is selected based on validation accuracy. Held-out test accuracy is measured through the benchmark evaluation service from the generated 400-image prediction file, because test labels are not released.

3.7 Compute Environment

Experiments are conducted using PyTorch 2.11.0 with CUDA 12.8 on an NVIDIA RTX 5070 Ti Laptop GPU (12 GB VRAM). The Python environment uses Python 3.12 with torchvision 0.26.0. Training hyperparameters are tuned per architecture: image sizes range from 160 to 288, batch sizes from 24 to 256 depending on model capacity and GPU memory constraints, and all models are trained with early stopping based on validation accuracy.

4 Experimental Results

4.1 Baseline Results

Our baseline is a ResNet18 model with light augmentation, no MixUp, and standard cross-entropy—a configuration closely matching the simplest transfer-learning setup. Table 4 reports the validation metrics.

4.2 Multi-Architecture Evaluation

Table 5 presents the validation accuracy and macro-F1 for evaluated architectures with complete local metric files. Figure 1 visualizes the performance comparison across architectures.

Table 4: ResNet18 baseline validation metrics. The model performs best on the visually more separable Small class.

Class	Precision	Recall	F1
Large	79.2%	75.0%	77.0%
Normal	75.6%	71.4%	73.4%
Small	73.8%	93.9%	82.7%
Overall: Accuracy=76.5%, Macro-F1=77.7%			

Table 5: Validation performance across locally evaluated architectures. Accuracy is reported with 95% Wilson confidence intervals using the 200-image validation split.

Architecture	Image Size	Valid Acc. (95% CI)	Macro-F1
Custom Residual CNN	160	67.0% [60.2, 73.1]	69.2%
ResNet18 (baseline)	224	76.5% [70.2, 81.8]	77.7%
ResNet18 rectangular	256×344	74.5% [68.0, 80.0]	76.2%
EfficientNetV2-S	224	73.5% [67.0, 79.1]	75.4%
ConvNeXt Tiny (no MixUp)	224	76.5% [70.2, 81.8]	77.8%
ConvNeXt Tiny (best)	224	77.5% [71.2%, 82.7%]	79.4%

4.3 Final Model Performance

Figure 2 visualizes the training trajectory of the final ConvNeXt Tiny model. Figure 3 presents the validation confusion matrix.

Key observations from the confusion matrix:

- **Large-Normal confusion** is the dominant error mode, accounting for most validation mistakes (36 out of 45 total errors). The heart size boundary between Large and Normal categories is inherently ambiguous.
- **Normal-Small confusion** accounts for the remaining errors ($8 + 1 = 9$), substantially lower than Large-Normal confusion.
- **Small recall** is excellent at 97.0%, with no Large-Small confusion whatsoever.

Table 6 presents the per-class metrics for the final ConvNeXt Tiny model.

4.4 Statistical Analysis

Because the validation split contains only 200 images, small differences between models should be interpreted

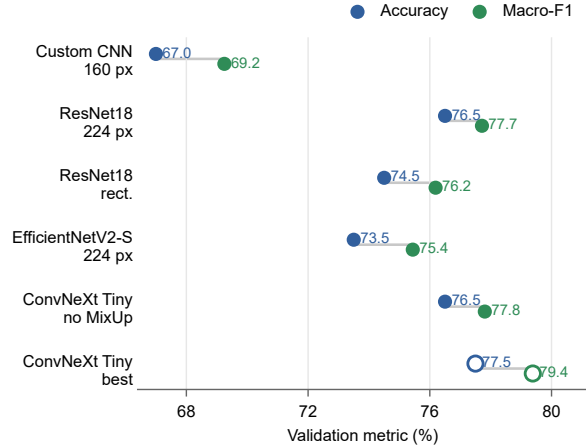


Figure 1: Validation accuracy and macro-F1 point estimates across architectures; Wilson confidence intervals for accuracy are reported in Table 5. The point-based comparison avoids over-emphasizing small differences while showing that the best ConvNeXt Tiny achieves 77.5% accuracy and 79.4% macro-F1. The improvement over the ResNet18 baseline (+1.0pp) does not reach statistical significance ($p = 0.856$ by McNemar’s test).

cautiously. The reported accuracy value of 77.5% corresponds to a 95% Wilson binomial confidence interval of approximately [71.2%, 82.7%] (reflecting binomial uncertainty for a single observation on $N = 200$ samples, not model variability across seeds or splits). Since checkpoint selection and hyperparameter tuning also use the validation split, validation metrics are optimistic estimates; the held-out test accuracy is the less selection-biased aggregate performance indicator available for this study.

We performed a paired McNemar exact test [6] on the fixed 200-image validation split comparing the final ConvNeXt-Tiny model against the ResNet18 baseline. The test was constructed from a 2×2 contingency table of paired predictions: $b = 14$ samples where ResNet18 was correct but ConvNeXt-Tiny was wrong, and $c = 16$ samples where ResNet18 was wrong but ConvNeXt-Tiny was correct. The two-sided exact binomial test yields $p = 0.856$, which does *not* indicate a statistically significant improvement over the baseline at conventional

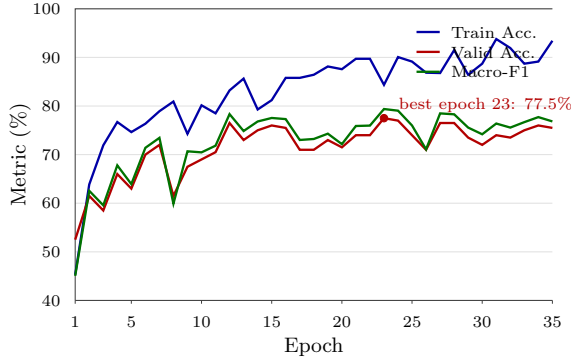


Figure 2: Training dynamics of the final ConvNeXt Tiny model trained at 224×224 resolution with strong augmentation, MixUp ($\alpha = 0.4$), label smoothing ($\varepsilon = 0.1$), and class-balanced loss. The best checkpoint occurs at epoch 23.

Table 6: Per-class validation metrics for the final ConvNeXt Tiny model.

Class	Precision	Recall	F1
Large	78.6%	72.4%	75.3%
Normal	75.6%	74.7%	75.1%
Small	80.0%	97.0%	87.7%
Overall: Accuracy=77.5%, Macro-F1=79.4%			

thresholds ($\alpha = 0.05$). The final-vs-no-MixUp comparison, recomputed from the corresponding per-image validation predictions, coincidentally yields the same discordant counts ($b = 14$, $c = 16$; $p = 0.856$). We note that the validation split was also used for model selection (hyperparameter tuning and checkpoint selection), which introduces a circularity concern: the same data influenced both which model was selected and the comparison between models. Because the validation set informed early stopping and hyperparameter decisions, the true Type-I error rate for this McNemar test may be higher than the nominal p -value; the reported $p = 0.856$ should therefore be interpreted cautiously under these conditions. Ideally, a significance test should be conducted on the held-out test set, but class-wise test labels are unavailable to us.

	Predicted label		
	Large	Normal	Small
Large	55	21	0
Normal	15	68	8
Small	0	1	32

Accuracy=77.5%, Macro-F1=79.4%

Figure 3: Validation confusion matrix (raw counts, $N = 200$) for the final ConvNeXt Tiny model. Rows represent true labels and columns represent predicted labels. The dominant error mode is confusion between Large and Normal classes ($21 + 15 = 36$ errors), while Small is rarely confused with Large (0 errors).

A 1000-sample bootstrap gives a 95% interval of [73.6%, 84.5%] for the final model’s macro-F1. The test-set accuracy of 77.25% is slightly below the validation accuracy of 77.5% (-0.25pp), which is consistent with validation-guided early stopping and the absence of substantial overfitting, and well within the validation confidence interval.

4.5 Comparison with Published Results

Table 7 compares our result with published baselines on the DogHeart dataset. We emphasize that the literature rows are not local reruns and follow different reporting protocols (training duration, hyperparameter search, evaluation splits). Therefore, the table should be read as contextual positioning rather than a strictly controlled head-to-head benchmark.

Table 7: Comparison with published DogHeart results. Literature rows follow the cited papers’ protocols, which may differ from our local evaluation setting.

Method	Valid Acc.	Held-out Test Acc.
Simple CNN [2]	–	72.0%
RVT (key-point) [4]	85.0%	87.3%
Ours: ResNet18 baseline	76.5%	–
Ours: ConvNeXt Tiny (best)	77.5%	77.25%

4.6 Exploratory Design-Space Exploration

We conducted an extensive hyperparameter search across architectures, input resolutions (160–288), augmentation strengths, and regularization settings. Table 8 summarizes representative runs reconstructed from local checkpoints and training histories.

Table 8: Exploratory training configurations reconstructed from checkpoint metadata and training logs. These are single-seed runs and should not be interpreted as controlled causal comparisons.

Configuration	Valid Acc.	Macro-F1
Custom CNN, 160, standard aug	67.0%	69.2%
ResNet18, 224, seed 11	72.0%	73.0%
ResNet18, 224, aspect-preserving pad	72.0%	74.4%
ResNet18, 256×344 rectangular	74.5%	76.2%
ResNet18, 320, light aug	74.5%	76.5%
EfficientNetV2-S, 224, standard aug	73.5%	75.4%
ResNet50, 256, fast exploratory run	64.0%	67.4%
MobileNetV3-Large, 224, exploratory run	56.0%	60.5%
RegNetY-400MF, 224, exploratory run	65.0%	67.5%
ConvNeXt Tiny, 224, light aug, no MixUp	76.5%	77.8%
ConvNeXt Tiny, 224, strong + MixUp + balanced	77.5%	79.4%
ConvNeXt Tiny, 288, RandAugment + MixUp	71.5%	73.1%
ConvNeXt Small, 288, RandAugment + MixUp	68.5%	70.6%

5 Discussion

5.1 Interpretation of Results

Our experiments demonstrate that a well-regularized ConvNeXt Tiny can achieve 77.5% validation accuracy and 79.4% macro-F1 on the DogHeart benchmark. However, the improvement over the ResNet18 baseline (76.5%) is only +1.0 percentage point and is not statistically significant (McNemar $p = 0.856$). This is consistent with the

small validation set size (200 images), which provides limited statistical power for detecting modest improvements.

The held-out internal test accuracy of 77.25% provides an aggregate check beyond the validation split, though we note that class-wise test metrics are unavailable because the test labels are not released.

5.2 Why Larger Models Did Not Always Improve

Several patterns from the exploratory runs merit discussion. Increasing image size to 288 or switching to larger ConvNeXt variants degraded validation performance, likely because the 200-image validation split makes model selection noisy and larger models overfit quickly. The best setting combined a moderate 224×224 input, strong but not excessive augmentation, MixUp, label smoothing, and class-balanced loss. At the same time, within-architecture seed/configuration variation is substantial (for example, ResNet18 ranges from 72.0% to 76.5% across listed runs), exceeding the +1.0pp margin between the final ConvNeXt Tiny and the baseline. Therefore, the ranking among local classification baselines should be interpreted as exploratory rather than definitive. The low ResNet50, MobileNetV3, and RegNetY results likely reflect under-tuning or architecture-setting mismatch rather than architectural inferiority.

5.3 Relation to Prior Work

Compared with the simple CNN DogHeart result of 72.0% [2], our best model improves held-out accuracy by 5.25 percentage points. However, our classification-only pipeline remains substantially below RVT’s key-point-supervised performance (87.3% test accuracy [4]). The gap between our 77.25% and RVT’s 87.3% highlights the value of anatomical supervision via key-point regression.

5.4 Limitations

Several important limitations apply:

- **Small validation set.** With only 200 validation images, performance estimates have wide confidence intervals ($\approx \pm 6\%$ at 95% level). The McNemar test

lacks power to detect small but potentially meaningful improvements.

- **Single-seed main result.** The final ConvNeXt Tiny checkpoint is from a single training run. Multi-seed evaluation would provide more reliable estimates of variance.
- **Exploratory ablations.** The design-space exploration was not conducted as a controlled factorial experiment, so individual component contributions cannot be isolated.
- **Architecture–hyperparameter confounding.** Architecture, image resolution, batch size, augmentation strength, and regularization settings vary together across runs, so the study cannot disentangle architecture effects from tuning effort.
- **No external-site validation.** Generalization beyond the original DogHeart dataset distribution is unknown.
- **No anatomical interpretability.** Unlike RVT, which predicts VHS key points, our model outputs only class probabilities without anatomical grounding.
- **Held-out test label limitations.** Only aggregate held-out test accuracy is available; class-wise test metrics cannot be computed because the labels are not released.
- **Possible data leakage.** Without patient-level metadata, we cannot verify that the same dog does not appear in both train and validation splits under different image IDs.

5.5 Future Work

For this specific task, the most promising next step is to inject anatomical information. Grad-CAM [7] could verify whether the classifier attends to the cardiac silhouette rather than dataset artifacts. A stronger extension would train a multi-head model that jointly predicts heart-size class and VHS landmarks, combining the simplicity of

classification backbones with the interpretability of RVT-style anatomical supervision. Multi-seed factorial experiments would also strengthen the claims by isolating the contributions of individual regularization components.

6 Conclusion

This paper presents a reproducible deep learning framework for canine heart-size classification on the DogHeart benchmark. Starting from a baseline pipeline (ResNet18 at 76.5%), we explore the interaction of architecture choice, data augmentation, and regularization under a shared evaluation framework. The best observed configuration—a ConvNeXt Tiny model with strong augmentation, MixUp, label smoothing, and class-balanced loss—achieves 77.5% validation accuracy, 79.4% validation macro-F1, and 77.25% on the held-out test set.

Several findings merit emphasis. First, the improvement over the ResNet18 baseline is modest (+1.0pp) and not statistically significant at conventional thresholds ($p = 0.856$), reflecting both the small validation set and the inherent difficulty of the Large-Normal boundary. Second, the dominant error mode is Large-Normal confusion, suggesting that the class boundary is intrinsically ambiguous at the radiographic appearance level. Third, our classification-only pipeline remains substantially below key-point-supervised RVT (87.3%), confirming that anatomical supervision is valuable for this task.

The complete codebase, trained model checkpoint, and generated prediction files are included to serve as a transparent and reproducible baseline for future work on DogHeart.

Data and Code Availability

The source code, training scripts, trained checkpoint, prediction CSV, environment requirements, checkpoint metadata, run histories, metric JSON files, and figure source-data CSV files are provided as supplementary materials accompanying this manuscript. The final ConvNeXt Tiny run uses random seed 20260525, as recorded in the checkpoint metadata. The DogHeart dataset is a public benchmark introduced in the RVT paper [4]; no additional animal data were collected for this study.

References

- [1] Ekin D. Cubuk, Barret Zoph, Jonathon Shlens, and Quoc V. Le. Randaugment: Practical automated data augmentation with a reduced search space. *Advances in Neural Information Processing Systems*, 33:18613–18624, 2020. 2, 3
- [2] Nikhil Deekonda. Assessing cardiomegaly in dogs using a simple cnn model. *arXiv preprint arXiv:2407.06092*, 2024. 1, 7
- [3] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 770–778, 2016. 2
- [4] Jialu Li and Youshan Zhang. Regressive vision transformer for dog cardiomegaly assessment. *Scientific Reports*, 14(1):1539, 2024. 1, 2, 7, 8
- [5] Zhuang Liu, Hanzi Mao, Chao-Yuan Wu, Christoph Feichtenhofer, Trevor Darrell, and Saining Xie. A convnet for the 2020s. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 11976–11986, 2022. 2
- [6] Quinn McNemar. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika*, 12(2):153–157, 1947. 5
- [7] Ramprasaath R. Selvaraju, Michael Cogswell, Abhishek Das, Ramakrishna Vedantam, Devi Parikh, and Dhruv Batra. Grad-cam: Visual explanations from deep networks via gradient-based localization. In *Proceedings of the IEEE International Conference on Computer Vision*, pages 618–626, 2017. 8
- [8] Christian Szegedy, Vincent Vanhoucke, Sergey Ioffe, Jonathon Shlens, and Zbigniew Wojna. Rethinking the inception architecture for computer vision. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 2818–2826, 2016. 3
- [9] Mingxing Tan and Quoc V. Le. Efficientnet: Rethinking model scaling for convolutional neural networks. In *International Conference on Machine Learning*, pages 6105–6114, 2019. 2
- [10] Hongyi Zhang, Moustapha Cisse, Yann N. Dauphin, and David Lopez-Paz. Mixup: Beyond empirical risk minimization. In *International Conference on Learning Representations*, 2018. 3